Artificial Intelligence Teaching Guide

Artificial intelligence technology has become increasingly sophisticated and readily available. We believe that educators can contribute to how this important technology is understood and used. We invite you to engage thoughtfully and attentively with this teaching guide as a way to learn about and positively influence the dialogue around artificial intelligence in education.

For instructors and teaching teams

We offer this guide to all instructors and teaching teams approaching the topic of generative AI tools in education, whether for the first time or as part of your ongoing engagement with the topic, in response to practical concerns that we heard from instructors like yourself. You don't need to be an expert or have prior experience with generative AI to use this resource, though you should have some understanding of or experience with teaching and learning in higher education contexts. We intend this guide to apply to

any disciplinary

area or teaching modality and to help you structure the work of integrating AI tools into your teaching practice.

Goals and scope of this guide

The goal of this guide is to help you make informed and intentional decisions as you navigate AI technology in your teaching practice and courses.

We cannot comprehensively address the complex topic of artificial intelligence in any short guide. Many campus service providers, such as University Information Technology (UIT), Stanford Accelerator for Learning, and the Institute for Human-Centered Artificial Intelligence (HAI), have developed excellent resources that offer

insight into

AI in terms of technical aspects, innovative new tools, societal impacts, AI research, and so on. We have chosen to focus on the practical and pedagogical aspects of AI tools in the classroom. We will focus on generative AI chatbots in particular, but you may find the content here can also apply to other generative AI tools, such as image, media, or code generators.

Multiple instructional modules for flexibility

Each page of this guide contains one instructional module, including content, practice tasks, and assessment activities. We suggest that you complete the activities and suggested readings

in each

section as a self-directed online lesson. We designed each module as a discrete and complete lesson that you can finish in a relatively short amount of time. You can work through the modules in any order. We encourage you to engage fully with each module, completing the recommended activities to reinforce your learning.

Each module has specific goals and objectives:

1. **Understanding AI literacy** — Examine a framework that identifies and organizes skills and knowledge useful for navigating generative AI.
2. **Defining AI and chatbots** — Define common concepts and explain how AI tools work
3. **Exploring the pedagogical uses of**

AI chatbots — Explore educational use cases, describe risks, and access and practice using chatbots.
4. **Analyzing the implications for your course** — Describe campus AI policy guidance, evaluate, and analyze your course
5. **Creating your course policy on AI** — Draft a course policy on AI use for your syllabus
6. **Integrating AI into assignments** — Examine ways to integrate AI tools into assignments and activities used to assess student learning

Complete the modules with your colleagues

Learning together with others can deepen the learning experience. We encourage you to organize your colleagues to complete these modules together.

Facilitate an AI workshop based on these modules

Stanford's Center for Teaching and Learning has also developed **do-it-yourself workshop kits** inspired by these modules. The kits expand upon the topics and strategies covered in these modules. Each workshop kit typically contains a resource list, sample agenda, promotional materials, slide presentation, facilitator's notes, key strategies, and an evaluation tool to assess learning and gather feedback.

- **Exploring Pedagogic Uses of AI Workshop Kit**
- **Analyzing the Implications of AI Workshop Kit**
- **Creating an AI Course Policy Workshop Kit**
- **Integrating AI into Assignments Workshop Kit**

Complete the self-paced Canvas resource on critical AI literacy

Stanford's Center for Teaching and Learning has also developed a **Critical AI Literacy for Instructors** self-paced, professional development resource available in Canvas. It is designed for instructors new to generative AI (genAI) technology and aims to provide foundational knowledge and skills to critically and effectively navigate genAI in teaching and learning contexts.

How AI was used in the creation of this guide

We did not copy and paste any language generated by AI chatbots into this guide.

We used

AI chatbots, primarily ChatGPT, to generate feedback on the clarity and structure of some of the writing and to clean up some text formatting. We used ChatGPT and other chatbots more extensively in the development of the module "Exploring the pedagogical uses of AI chatbots" to mimic how we thought instructors and students might use them in a course and to better understand the pedagogical potential and challenges of such tools.

Authors and acknowledgments

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Stanford Teaching Commons. If you have any questions, contact us at TeachingCommons@stanford.edu.

Thank you!

Preview of the first module

Understanding AI literacy

An AI literacy framework that identifies and organizes skills and knowledge useful for navigating generative AI in education.

Go to the first module